

■ 1.8.7 Advanced Topic: Plotting Tables of Functions

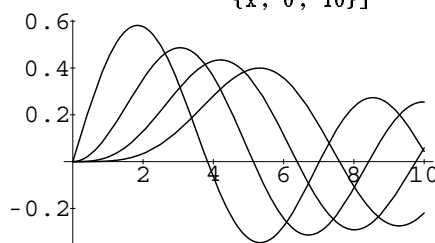
When you ask *Mathematica* to plot an object, say f , as a function of x , there are two possible approaches it can take. One approach is first to try and evaluate f , presumably getting a symbolic expression in terms of x , and then subsequently evaluate this expression numerically for the specific values of x needed in the plot. The second approach is first to work out what values of x are needed, and only subsequently to evaluate f with those values of x .

If you type `Plot[f, {x, xmin, xmax}]` it is the second of these approaches that is used. This has the advantage that *Mathematica* only tries to evaluate f for specific numerical values of x ; it does not matter whether sensible values are defined for f when x is symbolic.

There are, however, some cases in which you need *Mathematica* to evaluate f before it starts to make a plot. One common case is when f is a command that generates a table of functions. You want to have *Mathematica* first produce the table, and then evaluate the functions, rather than trying to produce the table afresh for each value of x . You can do this by typing `Plot[Release[f], {x, xmin, xmax}]`. The `Release` tells *Mathematica* not to “hold” f unevaluated.

This makes a plot of the Bessel functions $J_n(x)$ with n running from 1 to 4. The `Release` tells *Mathematica* first to make the table of functions, and only then to evaluate them for particular values of x .

```
In[1]:= Plot[Release[Table[BesselJ[n, x], {n, 4}]],
             {x, 0, 10}]
```



<code>Plot[f, {x, xmin, xmax}]</code>	<i>first choose specific numerical values for x, then evaluate f for each value of x</i>
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<code>Plot[Release[f], {x, xmin, xmax}]</code>	<i>first evaluate f, then choose specific numerical values of x</i>
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<code>Plot[Release[Table[f, ..]], {x, xmin, xmax}]</code>	<i>generate a list of functions, and then plot them</i>
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Methods for setting up objects to plot.

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The issue of what order to use in choosing x , and in evaluating f , arises not only in `Plot`, but also in other iteration functions, such as `Sum` and `Table`. Section 2.2.19 discusses the general principles of evaluation for such functions.